PROJECT REPORT

Sales Forecast Preparation

Day 11 Internship Project

Project Focus	Preparing historical sales data as a monthly time series for forecasting analysis.
Tools	Python, Pandas, Matplotlib, Google Colab, Microsoft Excel
Dataset	Sample Superstore sales dataset (CSV format)
Deliverables	Prepared monthly sales CSV and sales trend chart

Prepared as a project documentation template. Add your measured results and chart before final submission.

1. Executive Summary

This project prepares transaction-level retail sales data for time-series analysis. The workflow loads the Superstore CSV into Pandas, converts the order date and sales fields to appropriate data types, removes records with invalid date or sales values, aggregates sales into monthly totals, and produces a line chart showing the historical sales pattern. The resulting dataset can support later forecasting work, but this preparation stage does not itself train or evaluate a forecasting model.

2. Project Objective

The objective is to create a clean, chronologically ordered monthly sales time series suitable for further forecasting analysis, while identifying potential data-quality issues such as invalid records and missing periods.

3. Dataset Description

The project uses a Sample Superstore CSV file. The workflow expects an order-date field named **Order Date** and a numeric sales field named **Sales**. Other columns may be present, but the monthly time series is built from these two fields. Confirm the exact dataset version and its date range from the output of your notebook.

4. Tools and Technologies

- Python — workflow and data processing
- Pandas — parsing, cleaning, sorting, and monthly aggregation
- Matplotlib — trend visualization
- Google Colab — cloud notebook environment
- Microsoft Excel — optional inspection of the exported CSV

5. Methodology

Step 1: Load the CSV file into a Pandas DataFrame.
Step 2: Standardize column names by removing leading and trailing spaces.
Step 3: Convert Order Date to datetime and Sales to numeric values.
Step 4: Remove rows where either required value is invalid or missing.
Step 5: Sort records by date and resample sales into monthly totals using month-start timestamps.
Step 6: Reindex to a continuous monthly frequency and apply the selected missing-month treatment.
Step 7: Export the prepared time series to CSV and create a monthly trend chart.

6. Data Quality and Assumptions

Invalid date and sales rows are excluded from the monthly aggregation. If a month is absent from the source data, filling it with zero assumes that no sales were recorded in that month. This may be incorrect if records are incomplete; verify missing months against the source before relying on them for forecasting. The count of zero-sales months should therefore be interpreted as a diagnostic, not automatically as proof of missing data.

7. Deliverables

- **prepared_sales_timeseries.csv** — monthly totals with Month and Total Sales columns.
- **sales_trend.png** — line chart of monthly sales.
- **sales_forecast.py** or the exported Colab notebook (.ipynb) — reproducible analysis code.
- **README.md** — setup instructions and project summary.

8. Results and Interpretation

Use your actual notebook output to complete this section. Record the number of source rows retained after cleaning, the first and last month, the number of monthly periods, total sales, and any zero-sales months. Describe whether the chart shows an upward or downward pattern, recurring peaks or troughs, or irregular fluctuations. Do not claim a statistically validated forecast unless a forecasting model has been built and tested.

9. Results Entry Table

Fill this table from the output printed by your Google Colab notebook.

Metric	Your result
Rows after cleaning	_____
First month	_____
Last month	_____
Number of monthly periods	_____
Total sales	_____
Zero-sales months to verify	_____

10. Conclusion

The project transforms raw transaction data into a structured monthly sales time series and visualizes historical sales behavior. This provides a foundation for future forecasting. Before modeling, confirm data completeness, investigate zero-sales periods, and select an appropriate validation strategy for time-ordered data.

11. Suggested Next Steps

• Verify the exported CSV against the original source data.
• Investigate missing or zero-sales months before forecasting.
• Consider a baseline forecast and compare models using a chronological train/test split.
• Track suitable error metrics, such as MAE or RMSE, on held-out future periods.
• Upload the notebook, output CSV, chart, and README to the internship GitHub repository.

Report note: This document is based on the workflow described for the project. Dataset-specific findings are intentionally left for you to enter from your actual Colab results.